Project Title:

Real-Time Network Traffic Monitoring & Packet Analysis

Done By:
C. Pravin Sai
2420030777
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Why Network Monitoring Matters

Modern networks generate **vast streams of data** every second. Without continuous visibility, performance degradation and security threats can go undetected — often until it is too late.

Project Goal

Demonstrate how **Kali Linux native tools** can be used to monitor live traffic and analyse packets dynamically — no enterprise hardware required.

Performance Visibility

Track bandwidth usage and identify congestion in real time

Security Awareness

Detect suspicious packet patterns before they escalate

Practical Tooling

Kali Linux provides powerful, accessible tools for live analysis

The Challenge

Problem Statement

🌊 Traffic Volume

Large volumes of network traffic are impractical to analyse manually

🚦 Congestion

Network congestion degrades performance with no clear root-cause signal

👁️ Blind Spots

Suspicious or anomalous traffic often goes completely unnoticed

📉 No Real-Time View

Lack of live visibility means problems are discovered too late

ⓘ **Core Question:** How can we monitor and analyse network traffic in real time using simple, accessible tools?

What We Set Out to Do

Project Objectives

1

Bandwidth Monitoring

Track real-time bandwidth usage across the network interface

2

Packet Capture

Intercept and log packets at the data-link layer

3

Device Communication

Identify active communication paths between networked devices

4

Dynamic Visualisation

Display traffic flow visually as it happens

5

Behavioural Insight

Build understanding of typical and atypical network behaviour

The Stack

Tools & Technologies

Kali Linux

Security-focused OS providing the complete environment for all tools used in this project

nload

Real-time terminal bandwidth visualiser — displays incoming and outgoing traffic as live graphs

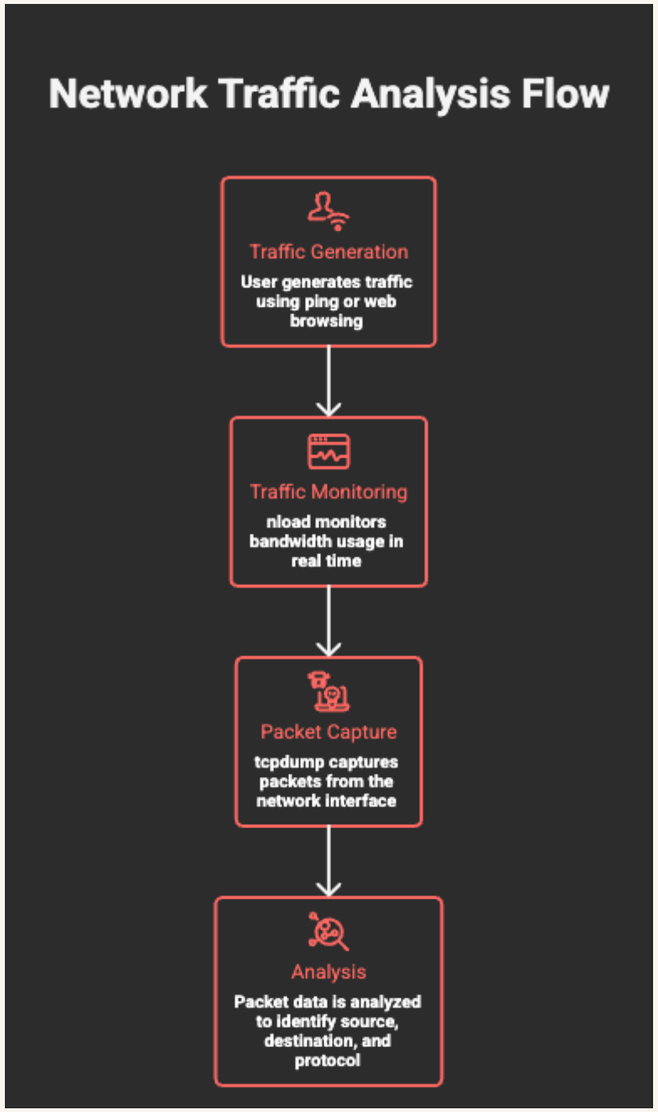
tcpdump

Command-line packet analyser that captures and inspects raw packets at the interface level

ping

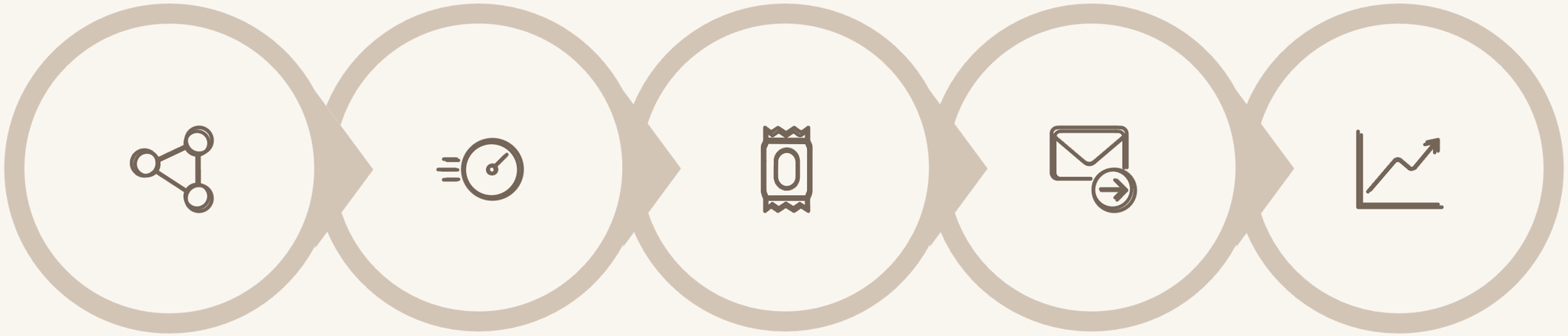
ICMP-based utility used to generate controlled traffic for live capture and observation

System Architecture:



Implementation

Methodology — Step by Step

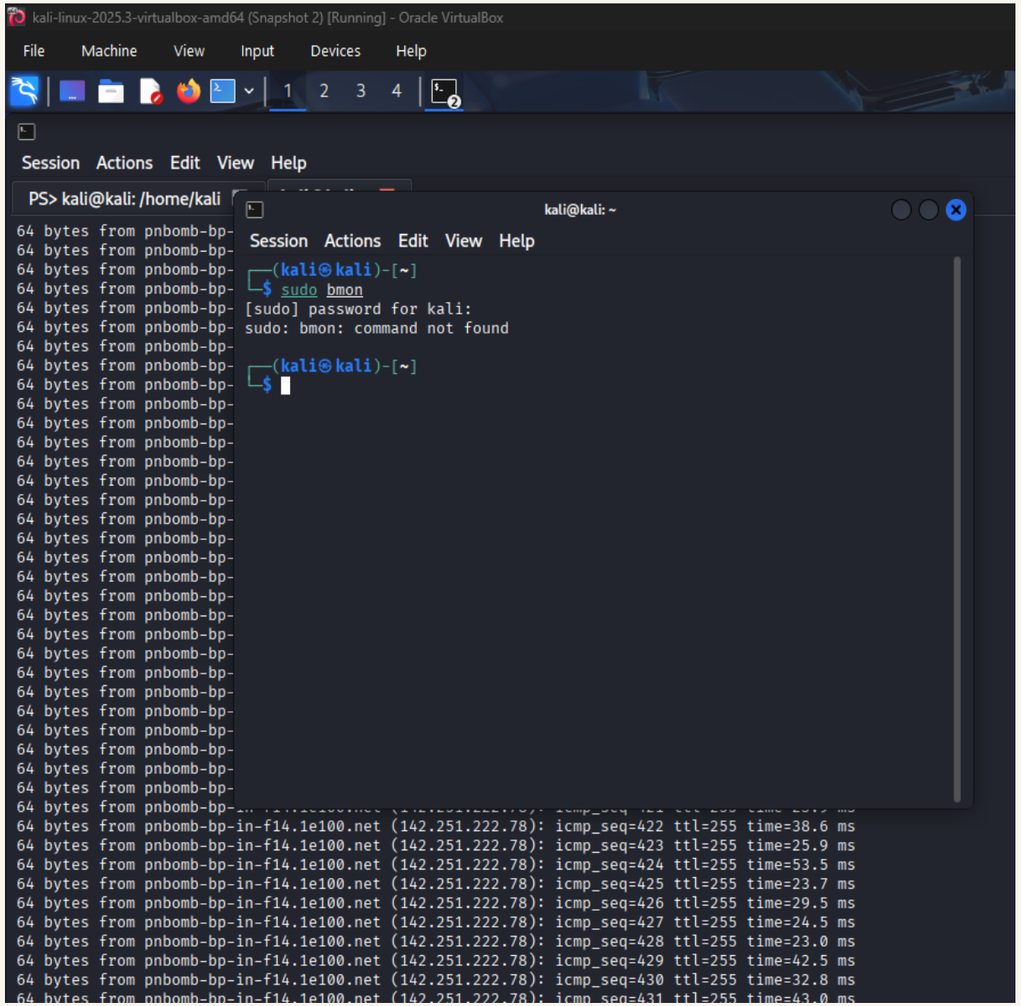


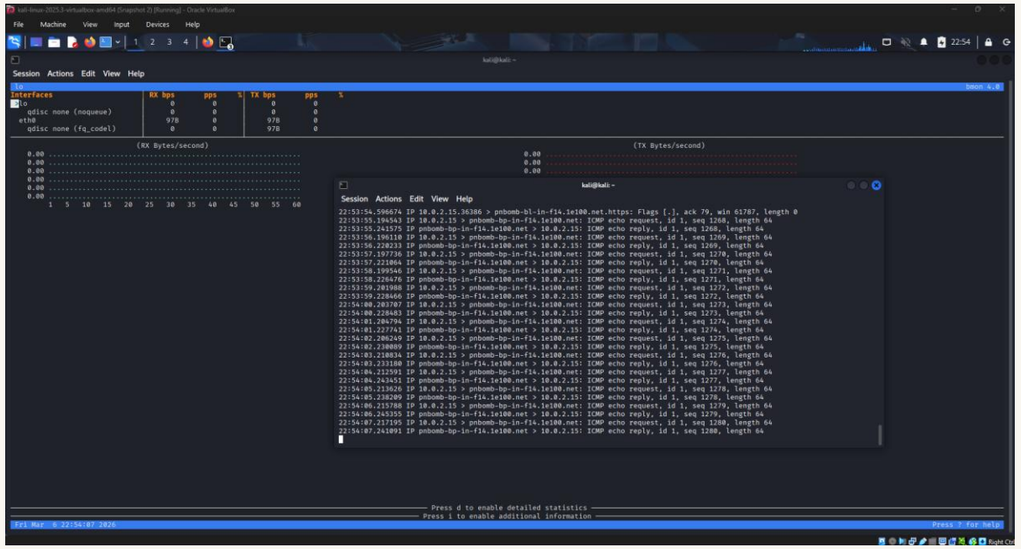
Identify Monitor Capture Generate Analyse

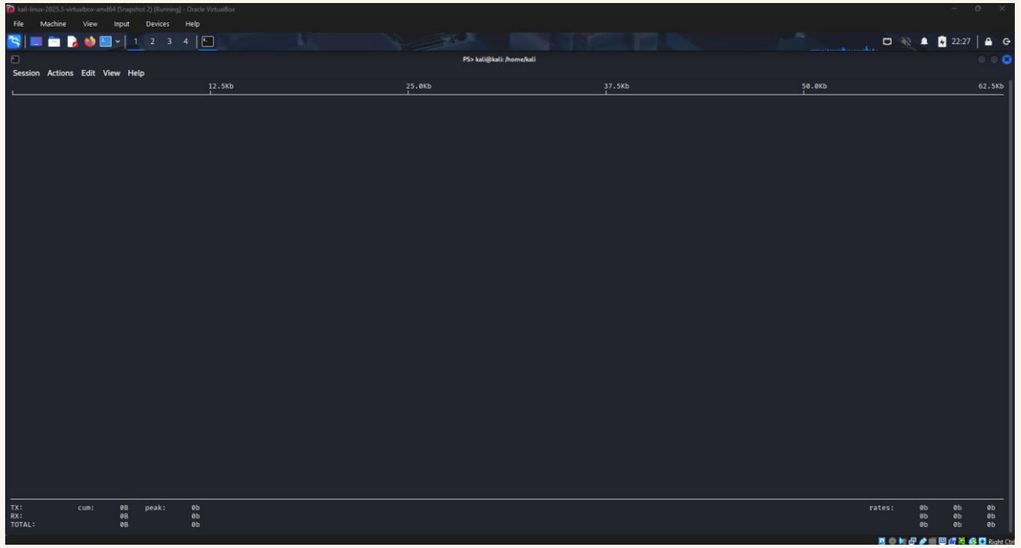
Interface Discovery: `ip a` lists all available network interfaces so the correct one is targeted

Packet Capture: `tcpdump -i <interface>` captures all passing packets in real time for immediate inspection

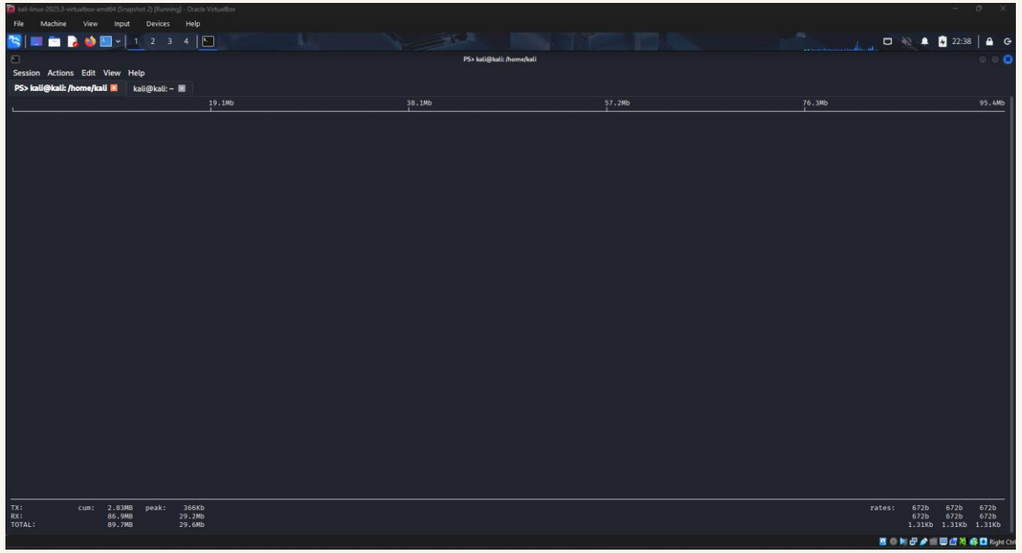
Results

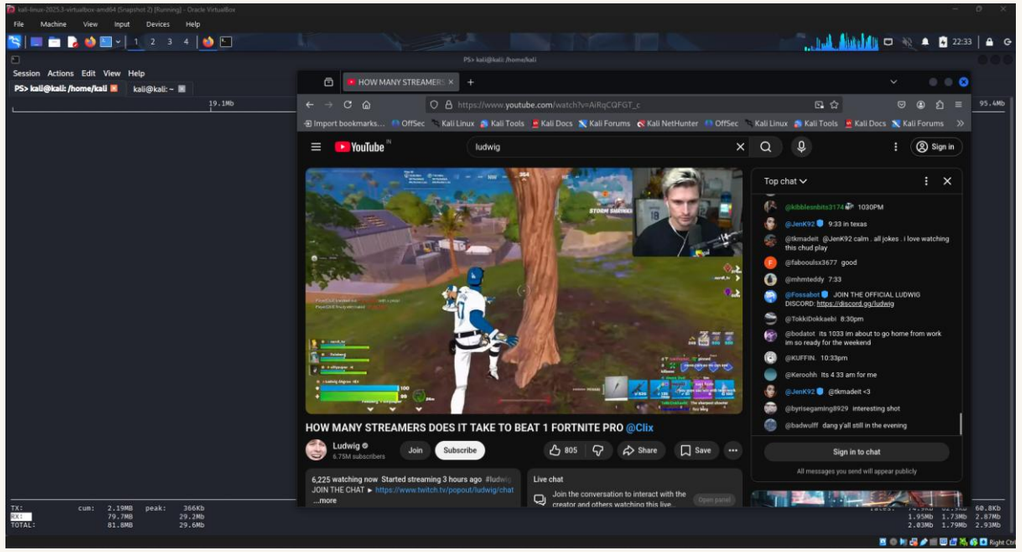

kali-linux-2025.3-virtualbox-amd64 (Snapshot 2) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Session Actions Edit View Help
PS> kali@kali: /home/kali
kali@kali: ~
Session Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ sudo bmon
[sudo] password for kali:
sudo: bmon: command not found
┌──(kali㉿kali)-[~]
└─$
64 bytes from pnbomb-bp-in-f14.1e100.net (142.251.222.78): icmp_seq=422 ttl=255 time=38.6 ms
64 bytes from pnbomb-bp-in-f14.1e100.net (142.251.222.78): icmp_seq=423 ttl=255 time=25.9 ms
64 bytes from pnbomb-bp-in-f14.1e100.net (142.251.222.78): icmp_seq=424 ttl=255 time=53.5 ms
64 bytes from pnbomb-bp-in-f14.1e100.net (142.251.222.78): icmp_seq=425 ttl=255 time=23.7 ms
64 bytes from pnbomb-bp-in-f14.1e100.net (142.251.222.78): icmp_seq=426 ttl=255 time=29.5 ms
64 bytes from pnbomb-bp-in-f14.1e100.net (142.251.222.78): icmp_seq=427 ttl=255 time=24.5 ms
64 bytes from pnbomb-bp-in-f14.1e100.net (142.251.222.78): icmp_seq=428 ttl=255 time=23.0 ms
64 bytes from pnbomb-bp-in-f14.1e100.net (142.251.222.78): icmp_seq=429 ttl=255 time=42.5 ms
64 bytes from pnbomb-bp-in-f14.1e100.net (142.251.222.78): icmp_seq=430 ttl=255 time=32.8 ms





Conclusion





Key Takeaway

Simple, built-in Kali Linux tools are sufficient to achieve meaningful, real-time network monitoring - no expensive hardware or proprietary software required.

This project proves that **visibility and security** can be accessible to any network engineer or student with the right open-source toolkit.